«Университет Иннополис»
(АНО ВО «Университет Иннополис»)

ВЫПУСКНАЯ КВАЛИФИКАЦИОННАЯ РАБОТА
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по направлению подготовки
09.03.01 – «Информатика и вычислительная техника»

GRADUATION THESIS
(BACHELOR'S GRADUATE THESIS)
Field of Study
09.03.01 – «Computer Science»

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«Информатика и вычислительная техника»
Area of Specialization / Academic Program Title:
«Computer Science»

Тема / Topic	**Система уточняющих типов для языка Dart / Refinement type system for Dart**

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Abstract

My abstract starts from here.

Chapter 1

Introduction

1.1 Background & Relevance

Index out of bounds errors are a well known reason for critical bugs appearing in modern software, mainly occuring in places where the checks for proper indexing are either insufficient or lacking entirely. The most notable example of these errors is access to an element of array that doesn't exist at the index provided.

Since a lot of programming languages rely on the array as a commonly used data structure, it is easy to encounter such an error in any part of the developed software. The examples include but are not limited to color conversions [need citation 89584 changelog], pixel values [need citation 8] or handling of empty lists.

Unfortunately, due to the nature of how array accesses happen, these errors are quite difficult to notice until they reach runtime or sometimes even production environments completely unnoticed.

1.2 Problem Statement

Dart programming language is prone to index out of bounds errors. The current type system, though incressing robustness [citation dart language specs], does not currently support collection-length constraints. This makes developers rely on assumptions and runtime testing.

This paper attempts to resolve this issue by introducing a refinement type system to statically confirm safety of array access operations while adding little annotation burden to the developers.

Chapter 2

Literature Review

Chapter 3

Methodology

Referencing other chapters 2, 3, 4, 5 and 6

TABLE 3.1
Simulation Parameters

A	B
Parameter	**Value**
Number of vehicles	$\mid \mathcal{V} \mid$
Number of RSUs	$\mid \mathcal{U} \mid$
RSU coverage radius	150 m
V2V communication radius	30 m
Smart vehicle antenna height	1.5 m
RSU antenna height	25 m
Smart vehicle maximum speed	v_{max} m/s
Smart vehicle minimum speed	v_{min} m/s
Common smart vehicle cache capacities	$[50, 100, 150, 200, 250]$ mb
Common RSU cache capacities	$[5000, 1000, 1500, 2000, 2500]$ mb

A	B
Common backhaul rates	$[75, 100, 150]$ mb/s

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Chapter 4

Implementation

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Chapter 5

Evaluation and Discussion

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Chapter 6

Conclusion

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Appendix A

Extra Stuff

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

Appendix B

Even More Extra Stuff

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.